Homog Response w/dyadic expansion

$$\underline{x}(t) = \sum_{K=1}^{n} \underbrace{(\underline{\omega}_K^T \underline{x}_0)}_{\text{weights}} \underbrace{e^{\lambda_K t} \underline{v}_K}_{\text{modes}}$$

$\underline{v}_K$ - Right eigenvectors of A

$\underline{\omega}_K$ - Left eigenvectors of A

λ_K - Eigenvalues of A

$\underline{x}_0 = \underline{x}(0)$ initial cond'ns

$\underline{\omega}_K^T \underline{x}_0 = \underline{\omega}_K \cdot \underline{x}_0$ dot product of left eigenvectors w/ICs.

Forced Sol'ns

$$\dot{\underline{x}}(t) = A\underline{x}(t) + B\underline{u}(t)$$

$B = [\underline{b}_1\ \underline{b}_2\ \cdots\ \underline{b}_m]$ (col part'n)

$$B\underline{u}(t) = \sum_{j=1}^{m} \underline{b}_j u_j(t)$$

$$\dot{\underline{x}}(t) = A\underline{x}(t) + \sum_{j=1}^{m} \underline{b}_j u_j(t)$$

$\underline{b}_j$ describes how j^{th} input effects rate of change of $\underline{x}(t)$

$\Rightarrow$ See this more explicitly in sol'ns for $\underline{x}(t)$

$$\underline{x}(t) = e^{At}\underline{x}_0 + \int_0^t e^{A(t-\tau)} B\underline{u}(\tau)\,d\tau$$

$$= e^{At}\underline{x}_0 + \sum_{j=1}^{m} \int_0^t e^{A(t-\tau)} \underline{b}_j u_j(\tau)\,d\tau$$

$$= \sum_{K=1}^{n} (\underline{\omega}_K^T \underline{x}_0) e^{\lambda_K t} \underline{v}_K + \sum_{j=1}^{m} \sum_{K=1}^{n} \int_0^t \left(e^{\lambda_K (t-\tau)} \underline{v}_K \underline{\omega}_K^T\right) \underline{b}_j u_j(\tau)\,d\tau$$

Dot product again!

Hence

$$\underline{x}(t) = \sum_{K=1}^{n} e^{\lambda_K t} \underline{v}_K \{\cdot\}$$

where

$$\{\cdot\} = \underline{\omega}_K^T \underline{x}_0 + \sum_{j=1}^{m} (\underline{\omega}_K^T \underline{b}_j) \int_0^t e^{-\lambda_K \tau} u_j(\tau)\,d\tau$$

amount of Kth mode due to input j

"Strength" of impact of jth input on K^{th} mode.

$\underline{\omega}_K \perp \underline{b}_j \Leftrightarrow$ j^{th} input has NO impact on K^{th} mode.

Output

$$\underline{y}(t) = C\underline{x}(t) + D\underline{u}(t)$$

$$= C\left[\sum_{K=1}^{n} e^{\lambda_K t}\underline{v}_K\{\cdot\}\right] + D\underline{u}(t)$$

row partition

$$C = \begin{bmatrix} \underline{c}_1^T \\ \vdots \\ \underline{c}_p^T \end{bmatrix} \qquad D = \begin{bmatrix} \underline{d}_1^T \\ \vdots \\ \underline{d}_p^T \end{bmatrix}$$

Then

for $i = 1, \ldots, p$:

$$y_i(t) = \sum_{K=1}^{n} e^{\lambda_K t}\underline{c}_i^T\underline{v}_K\{\cdot\} + \underline{d}_i^T\underline{u}(t)$$

How much of Kth mode is "visible" in i^{th} output

if $\underline{c}_i \perp \underline{v}_K \Rightarrow K^{th}$ mode not visible in i^{th} output.

Let $z_K(t) = e^{\lambda_K t}\{\cdot\}$ in the above eq'ns

Then

1.) $\underline{x}(t) = \sum_{K=1}^{n} z_K(t)\underline{v}_K$ — $z_K(t)$ coords. of $\underline{x}(t)$ in basis given by $\{\underline{v}_K\}_{K=1}^{n}$

and

2.) $\dot{z}_K(t) = \frac{d}{dt}z_K(t)$

$$= \lambda_K \underbrace{e^{\lambda_K t}\{\cdot\}}_{=z_K(t)} + \underbrace{e^{\lambda_K t}\frac{d}{dt}\{\cdot\}}_{=\sum_{j=1}^{m}(\underline{w}_K^T\underline{b}_j)u_j(t)}$$

so that

$$\boxed{\dot{z}_K(t) = \lambda_K z_K(t) + \sum_{j=1}^{m}(\underline{w}_K^T\underline{b}_j)u_j(t)}$$

Decoupled!!

Laplace:

$$\Rightarrow z_K(s) = \frac{1}{(s-\lambda_K)}\sum_{j=1}^{m}(\underline{w}_K^T\underline{b}_j)U_j(s)$$

and

$$Y_i(s) = \sum_{K=1}^{n}(\underline{c}_i^T\underline{v}_K)z_K(s) + \sum_{j=1}^{m}d_{ij}U_j(s)$$

$G(s)$ satisfies

$$\underline{Y}(s) = G(s)\,\underline{U}(s)$$

(when $\underline{x}_0 = \underline{0}$)

$$\Rightarrow [G(s)]_{ij} = \frac{Y_i(s)}{U_j(s)}$$

substitute expressions above:

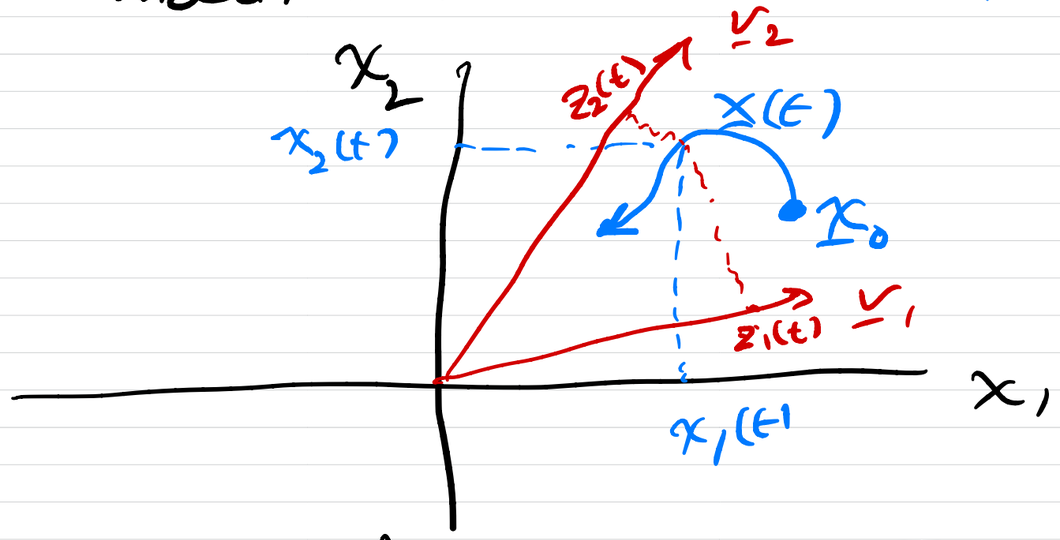
$$[G(s)]_{ij} = \sum_{k=1}^{n} \frac{(\underline{c}_i^T \underline{v}_k)(\underline{w}_k^T \underline{b}_j)}{(s-\lambda_k)} + d_{ij}$$

instantly available in partial fraction form!

$$= \underline{x}(t) = \sum_{k=1}^{n} z_k(t)\,\underline{v}_k$$

expresses sol'n in a new basis, given by eigenvectors of A.

This is sometimes called the "modal basis" and $z_k(t)$ the "modal coordinates".



$$\underline{x}(t) = \sum_{k=1}^{n} z_k(t)\,\underline{v}_k = V\,\underline{z}(t)$$

$$V = [\underline{v}_1 \cdots \underline{v}_n] \qquad \underline{z}(t) = \begin{bmatrix} z_1(t) \\ \vdots \\ z_n(t) \end{bmatrix}$$

Then

$$A\underline{x}(t) + B\underline{u}(t) = \dot{\underline{x}}(t) = V\,\dot{\underline{z}}(t)$$

$\underline{x}(t) = V\,\underline{z}(t)$

$\Lambda = \mathrm{diag}\{\lambda_k\}$

$$\Rightarrow \dot{\underline{z}}(t) = \underbrace{V^{-1}AV}_{\text{"}A_m\text{" "modal A"}}\underline{z}(t) + \underbrace{V^{-1}B}_{\text{"}B_m\text{" modal B}}\underline{u}(t)$$

$$\dot{\underline{z}} = A_m \underline{z} + B_m \underline{u}$$

$$A_m = V^{-1} A V = \Lambda$$

$$B_m = V^{-1} B$$

"modal" form of system dynamics

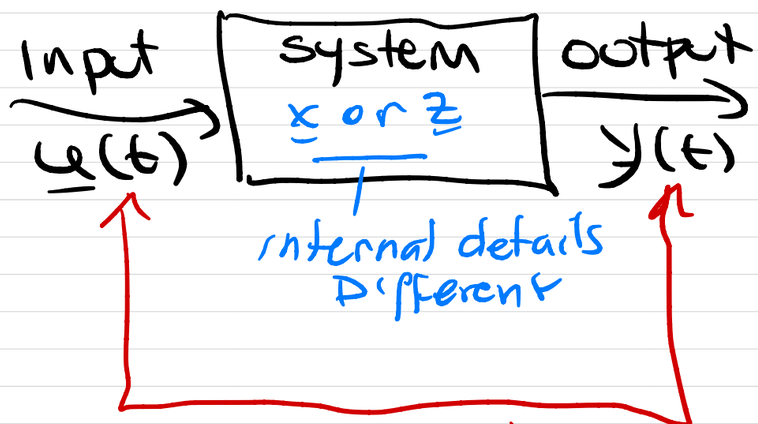
$$\underline{y} = C\underline{x} + D\underline{u}$$

$$= C(V\underline{z}) + D\underline{u}$$

$$\Rightarrow \quad \underline{y} = C_m \underline{z} + D_m \underline{u}$$

$$C_m = CV$$

$$D_m = D$$



same in both models

$\Rightarrow$ same input/output behavior.

Note: There is nothing special about Eigenvector matrix V in this approach!

Can do this with arbitrary, nonsingular P, i.e.

$$\underline{x}(t) = \sum_{k=1}^{n} z_k(t) P_k = P\underline{z}(t)$$

$$P = [P_1 \; P_2 \; \cdots \; P_n]$$

$\{P_k\}_{k=1}^{n}$ linearly indep.

$$\Rightarrow \quad \dot{\underline{z}}(t) = A_p \underline{z}(t) + B_p \underline{u}(t)$$

$$\underline{y}(t) = C_p \underline{z}(t) + D_p \underline{u}(t)$$

with $A_p = P^{-1} A P$, $B_p = P^{-1} B$

$C_p = CP$, $D_p = D$

This holds for a time-varying basis as well, i.e. $\{P_k(t)\}_{k=1}^{n}$ linearly indep. at each t.

Let $$P(t) = [P_1(t) \cdots P_n(t)]$$

and $$x(t) = \sum_{k=1}^{n} z_k(t) P_k(t)$$

$$= P(t) z(t)$$ new term

Then $$\dot{x}(t) = \dot{P}(t) z(t) + P(t) \dot{z}(t)$$

$$\Rightarrow \quad \dot{z}(t) = A_p(t) z(t) + B_p(t) u(t)$$
$$y(t) = C_p(t) z(t) + D_p(t) u(t)$$

where now $$A_p(t) = P^{-1}(t)[AP(t) - \dot{P}(t)]$$
$$B_p(t) = P^{-1}(t) B$$
$$C_p(t) = CP(t)$$
$$D_p(t) = D$$

Above is the most general form for coord. transformation. Note if $P(t)$ is in fact constant, then $\dot{P}(t) = 0$, and these eq'ns collapse to the previous versions $A_p = P^{-1}AP$

Also note, nothing changes here if A, B, etc are themselves time-varying (i.e.

$$A_p(t) = P^{-1}(t)[A(t)P(t) - \dot{P}(t)]$$
$$B_p(t) = P^{-1}(t) B(t)$$
$$C_p(t) = C(t)P(t)$$
$$D_p(t) = D(t)$$

Thus, the coord. change game can be played equally well with LTI/LTV systems, and constant or time-varying basis vectors for the new coords.